

DATA ANALYTICS – PROJECT 1 REPORT

Data Cleaning & Preparation

Submitted By:

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Organization:

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Project Title:

Data Cleaning & Preparation

Objective

The objective of this project was to clean and prepare a raw dataset by identifying missing values, checking duplicate records, correcting data formats, and ensuring data quality for further analysis.

Dataset Overview

The provided dataset contained customer order and transaction-related information. The dataset consisted of multiple fields including Order ID, Product Details, Quantity, Pricing Information, Payment Method, Order Status, Coupon Codes, and Dates.

Tasks Performed

1. Missing Value Analysis

The dataset was examined for missing or null values across all columns.

Findings:

- Missing values were found in the CouponCode column.
- A total of 309 missing entries were identified.

Action Taken:

- All blank CouponCode cells were replaced with "No Coupon" to maintain consistency and avoid null values.

2. Duplicate Record Verification

The dataset was checked for duplicate records and duplicate identifiers.

Findings:

- No duplicate rows were found.
- No duplicate Order IDs were found.
- No duplicate Tracking Numbers were found.

Action Taken:

- Dataset integrity was verified and no duplicate records required removal.

3. Date Format Standardization

Date values were reviewed to ensure consistency.

Action Taken:

- All dates were standardized to the format:
YYYY-MM-DD

This ensured uniformity and improved data quality.

4. Numeric Data Validation

Numeric fields such as Quantity, Unit Price, Items in Cart, and Total Price were examined.

Action Taken:

- Numeric values were verified.
- Decimal formatting was standardized where necessary.

5. Text Data Cleaning

Text-based columns were inspected for formatting inconsistencies.

Action Taken:

- Extra spaces were removed using Excel's TRIM function.
- Text formatting was reviewed for consistency.

Tools Used

- Microsoft Excel
- Excel Functions (TRIM, Filters, Remove Duplicates, Format Cells)

Results

After completing the data cleaning process:

- Missing values were successfully handled.
- Duplicate records were verified and eliminated where necessary.
- Date formats were standardized.
- Numeric formatting was validated.
- Text inconsistencies were corrected.

Final Dataset Status

- ✓ Zero Missing Values in CouponCode Column
- ✓ Zero Duplicate IDs
- ✓ Zero Duplicate Records
- ✓ Zero Incorrectly Formatted Dates
- ✓ Clean and Analysis-Ready Dataset

Conclusion

This project provided practical experience in data cleaning and preparation, which is a critical step in the data analytics process. Through this project, data quality issues such as missing values, formatting inconsistencies, and potential duplicate records were identified and resolved. The final dataset is clean, reliable, and ready for further analysis and reporting.